Your first PCB field simulation

A single microstrip trace in openEMS

This model is deliberately small: one copper trace, one FR-4-like dielectric block, one ground plane, and two ports. The solver calculates how an electromagnetic wave reflects from and passes through the trace.

Port 1 →	20 mm copper trace	→ Port 2
	assumed FR-4, εr = 4.2	
	ideal ground plane	

Run it

From the simulations directory, run:
`./scripts/run.sh`

Read the plot

S11 is the part of the signal reflected toward the source. Lower S11 means a better match to the 50 Ω port reference. **S21** is the part that reaches the other end. For this short, lossless line, S21 should stay near 0 dB.

Change one input at a time

In `scripts/run_microstrip.py`, change `TRACE_WIDTH_UM` (3000 µm), `SUBSTRATE_HEIGHT_UM` (1600 µm), or `SUBSTRATE_ER` (4.2). Run again and compare the new plot with the old one. The first change to try is trace width: wider trace usually lowers impedance and changes reflection.

Assumptions

The model uses ideal copper and a lossless dielectric. It is a learning example, not a prediction of the complete SRA board. Use the PCB fabricator's actual stackup values before treating a result as a design decision.